



Weekly Report

- **Title:** Fortnightly Training Report
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- **Program:** Bachelor of Computer Science
- **Week:** Week 7 and Week 8
- **Dates:** 1 Mar to 12 Mar 2026
- **Organization/Company Name:** Wadi Makkah Technology Company
- **Supervisor Name:** Mr. Khalid Abdulghani
- **Submission Date:** 13 Mar 2026

1. Introduction

This report documents the tasks completed during the seventh and eighth weeks of the cooperative training period, focusing on the development of a comprehensive restaurant management system. During this period, I worked on building the backend of the system using Node.js and Express.js, and creating RESTful APIs that support all core operations, from managing products and users to processing orders and invoices. In addition, an authentication and role-based authorization system was implemented to ensure system security and manage access according to user roles, with a focus on data integrity and operational stability through comprehensive system testing.

2. Tasks and Responsibilities

Week 7:

- **Project Structure and Backend Development:** we started developing the backend of the system using Node.js with the Express.js framework in order to build a set of RESTful APIs for managing the different operations of the restaurant system. We also created a new project and organized the file structure by dividing it into main folders such as models, controllers, routes, and middleware, which helped separate system components and made the code more organized and easier to maintain and extend in the future.

- **Database Integration and API Routes Implementation:** We connected the application to an SQLite database using Sequelize ORM. Several models were created to represent the main database tables, including User, Category, and Product, with attributes such as product name, price, and available stock quantity. After defining the models, we implemented the API routes for managing users and products and linked them to their corresponding controllers, which handle operations such as creating, updating, and retrieving data from the database.

- **Authentication and Authorization System:** We implemented an authentication system using JSON Web Token (JWT). A token is generated when a user logs in and is sent in the Authorization header with subsequent requests to verify the user's identity. In addition, we developed a middleware component to handle Role-Based Authorization, allowing the system to assign different permissions based on the user's role. For example, users with the Admin role can perform administrative operations such as creating categories, adding products, or updating product information like price and stock quantity.

Week 8:

This week, we continued developing the system's order management functionality.

- **Order Management and Models Creation:** We created additional models in the database, such as Order and Invoice, and linked them with products and users using Sequelize Associations to ensure proper relationships between the tables and smooth data flow within the system.

- **API Endpoint and Data Validation:** We developed an API Endpoint for creating orders, allowing users to send a list of products with the desired quantities. Within the Controller responsible for order creation, all necessary validations were performed to ensure that the products exist in the database and that the requested quantities do not exceed the available stock, ensuring accurate and reliable order data.

- **Total Price Calculation and Invoice Creation:** After successful validation, the system calculates the total price of the order and creates an associated Invoice record. The price is determined by multiplying the price of each product by the requested quantity and summing all values to obtain the total order amount.



- **Stock Update and Data Integrity:** The stock quantity for each product is updated immediately after the order is completed. All order-related operations, including creating the order, generating the invoice, and updating stock, are executed within a Database Transaction using Sequelize to ensure data integrity.

- **System Testing and Validation:** We tested all API Endpoints to verify that login, product management, order creation, and stock updates function correctly. These tests ensure that the system operates efficiently and meets the requirements of restaurant management through stable and effective APIs.

3. Skills and Knowledge Gained

- Developing comprehensive RESTful APIs using Node.js and Express.js.
 - Implementing authentication and role-based authorization using JSON Web Token (JWT).
 - Utilizing Database Transactions to ensure safe and reliable execution of complex operations.
 - Testing the system to ensure functional correctness and operational stability after executing multi-step processes.
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4. Challenges Faced

During the development of the system, I faced a challenge in properly defining user permissions and securing access to sensitive operations, especially those restricted to users with the Admin role. Initially, it was possible for any registered user to access certain protected resources if their permissions were not correctly verified.

5. Solutions and Improvements

I implemented an authentication system using JSON Web Token (JWT), along with middleware for role-based authorization. This setup allows the system to verify the user's identity through the token sent with each request and prevents unauthorized access to sensitive operations. After testing, I confirmed that all protected operations are now properly secured, and the system functions safely and efficiently.



6. Conclusion

During these two weeks, my experience in developing the system was filled practical opportunities. I learned how to design a secure authentication system and manage user permissions using JWT, while ensuring data integrity even during complex operations. Every challenge I faced became an opportunity to enhance my skills, turn problems into practical solutions, and make the learning experience both engaging and rewarding. I also aspire to learn modern technologies and methodologies that make systems more secure and efficient in management, expanding my ability to design integrated and stable systems in the future.

End of Report

